Chapter 1

Bayesian Filter

§ 1.1

Basic Settings

Let

$$\boldsymbol{x}_t \in \mathbb{R}^{n_{\boldsymbol{x}}}$$

denote the state vector at time t, which represents the unknow quantity we want to infer.

At each time t, we observe

$$\boldsymbol{z}_t \in \mathbb{R}^{n_{\boldsymbol{z}}}.$$

The observation contains information about the hidden state $\boldsymbol{x}_t$, but may be noisy and uncertain. For convenience, deine the observation history up to time t as

$$\boldsymbol{z}_{1:t} = \{\boldsymbol{z}_1, \ldots, \boldsymbol{z}_t\}$$

Similarly, suppose that a control input is available at each time t, denoted by

$$\boldsymbol{u}_t \in \mathbb{R}^{n_{\boldsymbol{u}}},$$

with the corresponding control history

$$\boldsymbol{u}_{1:t} = \{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_t\}.$$

Therefore, the available information evolves sequentially as

$$\boldsymbol{u}_1, \boldsymbol{z}_1, \boldsymbol{u}_2, \boldsymbol{z}_2, \boldsymbol{u}_3, \boldsymbol{z}_3, \ldots$$

whereas the states

$$\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots.$$

are not directly observed.

Since subsequent inference depends on the state, the objective is to characterize the uncertainty associated with the random variable $\boldsymbol{x}_t$ given all available information. This uncertainty is represented by the belief state

$$\mathrm{bel}(\boldsymbol{x}_t) = p(\boldsymbol{x}_t|\boldsymbol{z}_{1:t}, \boldsymbol{u}_{1:t}).$$

The belief state is therefore a probability distribution over the possible values of $\boldsymbol{x}_t$ conditioned on the available observation and control histories. It is also commonly referred to as the posterior state distribution, information state, or state of knowledge.

§ 1.2 *Core Formula of Bayesian Filter*

The key idea of the Bayesian filter is to represent uncertainty explicitly using the calculas of probability theory.

At each time step, the hidden state $\boldsymbol{x}_t$ evolves from the previous state $\boldsymbol{x}_{t-1}$ under the control input $\boldsymbol{u}_t$, while the observation $\boldsymbol{z}_t$ is generated from the corresponding state $\boldsymbol{x}_t$, as shown in Figure 1.1.

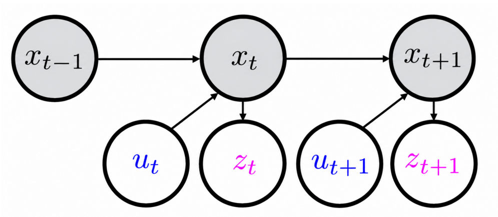

Figure 1.1 Graphical representation of the Bayesian filtering model

However, the belief, $p(\text{current sate}|\text{all past information})$, is intractable. This leads to the Markov assumption: future state conditionally independent of past control inputs and observations given present state. We therefore have

$$p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}, \boldsymbol{u}_t, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1}) = p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1}, \boldsymbol{u}_t),$$

$$p(\boldsymbol{z}_t|\boldsymbol{x}_t, \boldsymbol{u}_t, \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1}) = p(\boldsymbol{z}_t|\boldsymbol{x}_t).$$

Using Bayes' rule and the Markov assumptions,

$$\begin{aligned}
\mathrm{bel}(\boldsymbol{x}_t) &= p(\boldsymbol{x}_t|\boldsymbol{z}_{1:t},\boldsymbol{u}_{1:t}) = p(\boldsymbol{x}_t|\boldsymbol{z}_t,\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t}) \leftarrow \text{Definition of belief}\\
&= \frac{p(\boldsymbol{x}_t,\boldsymbol{z}_t|\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t})}{p(\boldsymbol{z}_t|\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t})} \leftarrow \text{Condition probability}\\
&= \eta_t p(\boldsymbol{x}_t,\boldsymbol{z}_t|\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t}) \leftarrow \text{Normalization constant}\\
&= \eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t,\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t})p(\boldsymbol{x}_t|\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t}) \leftarrow \text{Bayes rule}\\
&= \eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t)p(\boldsymbol{x}_t|\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t}) \leftarrow \text{Markov assumption}\\
&= \eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t)\int p(\boldsymbol{x}_t,\boldsymbol{x}_{t-1}|\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t})\mathrm{d}x_{t-1} \leftarrow \text{Marginalization}\\
&= \eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t)\int p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1},\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t})p(\boldsymbol{x}_{t-1}|\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t})\mathrm{d}x_{t-1} \leftarrow \text{Probability chain rule}\\
&= \eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t)\int p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1},\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t})p(\boldsymbol{x}_{t-1}|\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t-1})\mathrm{d}x_{t-1} \leftarrow \text{Unnecessary condition}\\
&= \underbrace{\eta_t p(\boldsymbol{z}_t|\boldsymbol{x}_t)}_{\text{correction}}\underbrace{\int p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1},\boldsymbol{u}_t)\mathrm{bel}(\boldsymbol{x}_{t-1})\mathrm{d}x_{t-1}}_{\text{prediction}}
\end{aligned}$$

where η_t is a normalization constant,

$$\eta_t^{-1} = p(\boldsymbol{z}_t|\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t}) = \int p(\boldsymbol{z}_t|\boldsymbol{x}_t)p(\boldsymbol{x}_t|\boldsymbol{z}_{1:t-1},\boldsymbol{u}_{1:t})\mathrm{d}\boldsymbol{x}_t$$

and $\mathrm{bel}(x_0) = p(\boldsymbol{x}_0)$.

The Bayesian filter therefore provides a recursive update approach to calculate belief tractably consisting of two steps: prediction, which propagates the previous belief through the state transition model, and correction, which incorporates the latest observation through the observation model.

§1.3 *Kalman Filter*

The Kalman filter is a special case of the Bayesian filter, in which the system is assumed to be linear and all uncertainties are modeled as Gaussian.

Suppose that the state transition $\boldsymbol{x}_t$ is linear in previous state $\boldsymbol{x}_{t-1}$ and control input $\boldsymbol{u}_t$, i.e.,

$$\boldsymbol{x}_t = \mathbf{F}_t\boldsymbol{x}_{t-1} + \mathbf{B}_t\boldsymbol{u}_t + \boldsymbol{q}_t, \quad \boldsymbol{q}_t \sim \mathcal{N}(\mathbf{0},\mathbf{Q}_t), \qquad \forall t \in \{1,2,\ldots,T\}.$$

Thus,

$$p(\boldsymbol{x}_t|\boldsymbol{x}_{t-1},\boldsymbol{u}_t) = \mathcal{N}(\mathbf{F}_t\boldsymbol{x}_{t-1} + \mathbf{B}_t\boldsymbol{u}_t,\mathbf{Q}_t)$$

Similarly, suppose that the observation model is also linear:

$$\boldsymbol{z}_t = \mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t, \quad \boldsymbol{r}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t), \qquad \forall t \in \{1, 2, \ldots, T\}.$$

Thus,

$$p(\boldsymbol{z}_t | \boldsymbol{x}_t) = \mathcal{N}(\mathbf{H}_t \boldsymbol{x}_t, \mathbf{R}_t)$$

Suppose that the posterior state distribution at time $t-1$ is Gaussian:

$$p(\boldsymbol{x}_{t-1} | \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t-1}) = \mathcal{N}(\boldsymbol{\mu}_{t-1|t-1}, \boldsymbol{\Sigma}_{t-1|t-1}),$$

where $\boldsymbol{\mu}_{t-1|t-1}$ and $\boldsymbol{\Sigma}_{t-1|t-1}$ denote the posterior mean and covariance of the state $\boldsymbol{x}_{t-1}$, respectively, based on all information available up to time $t-1$.

Because a linear transformation of a Gaussian random variable remains Gaussian, the predictive distribution is

$$p(\boldsymbol{x}_t | \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) = \mathcal{N}(\boldsymbol{\mu}_{t|t-1}, \boldsymbol{\Sigma}_{t|t-1}).$$

Its mean is

$$\begin{aligned}
\boldsymbol{\mu}_{t|t-1} &= \mathbb{E}[\boldsymbol{x}_t | \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}] \leftarrow \text{Definition of predictive mean} \\
&= \mathbb{E}[\mathbf{F}_t \boldsymbol{x}_{t-1} + \mathbf{B}_t \boldsymbol{u}_t + \boldsymbol{q}_t] \leftarrow \text{State transition model} \\
&= \mathbf{F}_t \mathbb{E}[\boldsymbol{x}_{t-1}] + \mathbf{B}_t \boldsymbol{u}_t + \mathbb{E}[\boldsymbol{q}_t] \leftarrow \text{Linearity of expectation} \\
&= \mathbf{F}_t \boldsymbol{\mu}_{t-1|t-1} + \mathbf{B}_t \boldsymbol{u}_t \leftarrow \text{Zero-mean process noise,}
\end{aligned}$$

and its covariance is

$$\begin{aligned}
\boldsymbol{\Sigma}_{t|t-1} &= \mathrm{CoV}(\boldsymbol{x}_t | \boldsymbol{z}_{1:t-1}, \boldsymbol{u}_{1:t}) \leftarrow \text{Definition of predictive covariance} \\
&= \mathrm{CoV}(\mathbf{F}_t \boldsymbol{x}_{t-1} + \mathbf{B}_t \boldsymbol{u}_t + \boldsymbol{q}_t) \leftarrow \text{State transition model} \\
&= \mathrm{CoV}(\mathbf{F}_t \boldsymbol{x}_{t-1} + \boldsymbol{q}_t) \leftarrow \text{Deterministic control input} \\
&= \mathrm{CoV}(\mathbf{F}_t \boldsymbol{x}_{t-1}) + \mathrm{CoV}(\boldsymbol{q}_t) + \mathrm{CoV}(\mathbf{F}_t \boldsymbol{x}_{t-1}, \boldsymbol{q}_t) + \mathrm{CoV}(\boldsymbol{q}_t, \mathbf{F}_t \boldsymbol{x}_{t-1}) \leftarrow \text{Covariance of a sum} \\
&= \mathrm{CoV}(\mathbf{F}_t \boldsymbol{x}_{t-1}) + \mathrm{CoV}(\boldsymbol{q}_t) \leftarrow \text{Independence of state and process noise} \\
&= \mathbf{F}_t \mathrm{CoV}(\boldsymbol{x}_{t-1}) \mathbf{F}_t^\top + \mathbf{Q}_t \leftarrow \text{Covariance under linear transformation} \\
&= \mathbf{F}_t \boldsymbol{\Sigma}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q}_t \leftarrow \text{Previous posterior covariance.}
\end{aligned}$$

For the observation model, the difference between the actual and predicted observations is

$$\boldsymbol{\nu}_t = \boldsymbol{z}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}.$$

Before observing $\boldsymbol{z}_t$, this prediction error has zero mean:

$$\begin{aligned}
\mathbb{E}[\boldsymbol{\nu}_t] &= \mathbb{E}[\mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}] \\
&= \mathbf{H}_t \mathbb{E}[\boldsymbol{x}_t] + \mathbb{E}[\boldsymbol{r}_t] - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1} \\
&= \mathbf{H}_t \boldsymbol{\mu}_{t|t-1} - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1} = 0
\end{aligned}$$

and covariance

$$\begin{aligned}
\mathrm{CoV}(\boldsymbol{\nu}_t) &= \mathrm{CoV}(\mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}) \\
&= \mathrm{CoV}(\mathbf{H}_t \boldsymbol{x}_t + \boldsymbol{r}_t) \\
&= \mathrm{CoV}(\mathbf{H}_t \boldsymbol{x}_t) + \mathrm{CoV}(\boldsymbol{r}_t) + \mathrm{CoV}(\boldsymbol{r}_t, \mathbf{H}_t \boldsymbol{x}_t) + \mathrm{CoV}(\mathbf{H}_t \boldsymbol{x}_t, \boldsymbol{r}_t) \\
&= \mathbf{H}_t \mathrm{CoV}(\boldsymbol{x}_t) \mathbf{H}_t^\top + \mathbf{R}_t \\
&= \underbrace{\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top}_{\text{uncertainty in predicted state}} + \underbrace{\mathbf{R}_t}_{\text{observation noise}} .
\end{aligned}$$

Since the predicted state and the observation are jointly Gaussian, conditioning on the new observation $\boldsymbol{z}_t$ gives the posterior state distribution

$$p(\boldsymbol{x}_t | \boldsymbol{z}_{1:t}, \boldsymbol{u}_{1:t}) = \mathcal{N}\left(\boldsymbol{\mu}_{t|t}, \boldsymbol{\Sigma}_{t|t}\right),$$

where

$$\begin{aligned}
\boldsymbol{\mu}_{t|t} &= \mathbb{E}[\boldsymbol{x}_t | \boldsymbol{z}_t] \leftarrow \text{Definition of conditional mean} \\
&= \boldsymbol{\mu}_{t|t-1} + \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{z}_t) \mathrm{Cov}(\boldsymbol{z}_t)^{-1} (\boldsymbol{z}_t - \mathbb{E}[\boldsymbol{z}_t]) \leftarrow \text{Conditional Gaussian formula} \\
&= \underbrace{\boldsymbol{\mu}_{t|t-1}}_{\text{predicted mean}} + \underbrace{\boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top \left(\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t\right)^{-1}}_{\text{uncertainty-dependent correction}} \underbrace{\left(\boldsymbol{z}_t - \mathbf{H}_t \boldsymbol{\mu}_{t|t-1}\right)}_{\text{observation prediction error}} \leftarrow \text{Substitution of mean and covariance.}
\end{aligned}$$

and

$$\begin{aligned}
\boldsymbol{\Sigma}_{t|t} &= \mathrm{Cov}(\boldsymbol{x}_t | \boldsymbol{z}_t) \leftarrow \text{Definition of conditional covariance} \\
&= \boldsymbol{\Sigma}_{t|t-1} - \mathrm{Cov}(\boldsymbol{x}_t, \boldsymbol{z}_t) \mathrm{Cov}(\boldsymbol{z}_t)^{-1} \mathrm{Cov}(\boldsymbol{z}_t, \boldsymbol{x}_t) \leftarrow \text{Conditional Gaussian formula} \\
&= \boldsymbol{\Sigma}_{t|t-1} - \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top \left(\mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R}_t\right)^{-1} \mathbf{H}_t \boldsymbol{\Sigma}_{t|t-1} \leftarrow \text{Substitution of covariance terms.}
\end{aligned}$$

Therefore, the new observation corrects the predicted state according to the discrepancy between the actual and predicted observations, while the magnitude of the correction depends on both the uncertainty of the state prediction and the observation noise.